

PATHOGENESIS

Mechanisms have been extensively studied in herpes-associated erythema multiforme (EM). It remains unclear whether similar pathophysiological processes apply to EM caused by other triggers.

Complete infectious herpes simplex virus (HSV) has never been isolated from lesions of herpes-associated EM. However, HSV DNA fragments—always including the viral polymerase (Pol) gene—have been detected in EM lesions using PCR assays. These fragments are found primarily in basal keratinocytes and the lower spinous layers of the epidermis. The HSV Pol protein is expressed in epidermal cells, leading to recruitment of HSV-specific T cells, including cytotoxic T lymphocytes. This virus-specific immune response is followed by a broader, non-specific inflammatory reaction mediated by autoreactive T cells. The resulting cytokine release contributes to the delayed hypersensitivity-like histopathology characteristic of EM lesions.

HSV is transiently present in the bloodstream around the time of symptomatic herpes recurrence. If keratinocytes were directly infected via hematogenous spread, widespread cutaneous herpes would be expected rather than the localized lesions seen in EM. Instead, evidence suggests that HSV DNA is transported to the skin by monocytes, macrophages, and particularly CD34⁺ Langerhans cell progenitors that express the cutaneous lymphocyte-associated antigen (CLA), a skin-homing receptor. Upregulation of adhesion molecules enhances the binding of these HSV-carrying mononuclear cells to vascular endothelium. Increased expression of HLA class I and adhesion molecules on endothelial cells likely contributes to the dermal inflammation observed in EM.

Upon reaching the epidermis, these cells transfer the viral Pol gene to keratinocytes. While the genetic material may persist for several months, Pol protein expression is transient, lasting only a few days—consistent with the self-limited nature of EM lesions.

Variations in the efficiency of viral DNA fragmentation, elevated levels of circulating CD34⁺ cells, or an exaggerated immune response to the Pol protein

may explain why only a small subset of individuals with recurrent herpes develop EM.

CLINICAL FINDINGS

Diagnosis begins with clinical suspicion for EM based on characteristic features. Skin biopsy and laboratory tests are primarily helpful when the diagnosis is uncertain.

The second step is to assess the need for hospitalization, particularly in cases of EM major (EMM) with oral involvement impairing feeding, suspected Stevens-Johnson syndrome (SJS), or severe systemic symptoms.

The third step involves identifying the underlying cause—most commonly recurrent herpes simplex infection, *Mycoplasma pneumoniae* infection, or other triggers—through patient history, chest imaging, or serological testing.

History

Prodromal symptoms are typically absent. When present, they are mild and resemble an upper respiratory tract infection (e.g., cough, rhinitis, low-grade fever).

In EMM, fever exceeding 38.5°C (101.3°F) occurs in approximately one-third of cases.

A history of prior EM episodes is reported in at least one-third of patients and supports the diagnosis. A detailed review of events within the preceding three weeks should focus on potential triggers, especially recurrent herpes outbreaks.

Cutaneous Lesions

The rash appears suddenly. In most patients, all lesions develop within three days, although some may experience successive crops during a single episode.

The number of lesions varies widely—from just a few to hundreds. They are typically symmetric and acral, predominantly affecting the extensor surfaces of the extremities (hands, feet, elbows, knees), face, and neck. Involvement of the thighs, buttocks, and trunk is less common.